Check for updates

Review Article

CT for Detection, Fluoroscopy for Confirmation: A Meta-Analysis of Imaging in Suspected Esophageal Perforation

The American Surgeon™
2026, Vol. 0(0) 1-10
© The Author(s) 2026
Article reuse guidelines:
sagepub.com/journals-permissions
DOI: 10.1177/00031348261472868
journals.sagepub.com/home/asu
Sage

Rachid Eduardo Noleto da Nobrega Oliveira, MD, MS[1], Lucas Monteiro Delgado[2], Leonardo Braga Gonçalves, MD[3], Marina Castello Branco Bastos[4], Felipe S. Passos, MD[5], Isabella Faria, MD[6], and Roman Petrov, MD[6]

Abstract

Background: Esophageal perforation is a time-critical emergency in which delayed diagnosis is associated with substantial morbidity and mortality. Fluoroscopic esophagography (FE) and computed tomography esophagram (CTE) are commonly used, but their optimal roles in the acute diagnostic pathway remain uncertain. We performed a systematic review and meta-analysis comparing the diagnostic performance of FE and CTE in adults with suspected esophageal perforation and clarifying their complementary utility.

Study Design: PubMed, Embase, and Cochrane Library were searched from inception to October 26, 2025, for observational studies evaluating both FE and CTE in adults with suspected esophageal perforation and providing data for 2 × 2 contingency tables. Pooled diagnostic estimates were calculated using univariate and bivariate random-effects models. Direct comparisons were performed using Z tests on transformed summary estimates.

Results: Four retrospective studies including 439 patients met inclusion criteria; 69 patients (15.6%) had confirmed perforation. FE showed pooled sensitivity of 69.2% (95% CI, 48.9-84.0) and specificity of 98.4% (95% CI, 96.3-99.4). CTE showed higher sensitivity (81.8%; 95% CI, 38.5-97.0) but lower specificity (86.7%; 95% CI, 67.3-95.4). Sensitivity and specificity did not differ significantly between modalities. However, FE had significantly higher PPV, PLR, and AUC than CTE, indicating stronger rule-in and overall discriminative performance.

Conclusions: CTE and FE should be considered complementary tests. CTE is well suited for initial evaluation because of its sensitivity and broader anatomic assessment, whereas FE provides stronger confirmatory value, supporting a sequential strategy of initial CTE followed by selective FE.

Keywords

esophageal perforation, computed tomography, fluoroscopic esophagography, diagnostic accuracy

Introduction

Esophageal perforation is a time-critical emergency with substantial morbidity and mortality when diagnosis is delayed, largely due to the rapid onset of mediastinitis, empyheama, and sepsis with ultimate patient demise without appropriate management.[1-3] Early recognition and timely intervention is crucial, including broad spectrum antibiotics, critical care support, and the choice between endoscopic or surgical source control.[4,5] In current practice, two imaging modalities are commonly used to diagnose esophageal perforation: contrast fluoroscopic esophagography (FE), which can demonstrate direct contrast extravasation, and computed tomography esophagram (CTE), which surveys the thoracoabdominal

[1]Department of Thoracic Surgery, Barretos Cancer Center, Barretos, Brazil
[2]Department of Medicine, Universidade Federal de Minas Gerais, Belo Horizonte, Brazil
[3]Department of Medicine, Universidade Federal de Uberlândia, Uberlândia, Brazil
[4]Department of Medicine, Universidade Federal de São João del Rei, São João del Rei, Brazil
[5]Department of Thoracic Surgery, MaterDei Hospital, Salvador, Brazil
[6]Department of Surgery, UTMB John Sealy School of Medicine, Galveston, TX, USA

Corresponding Author:
Rachid Eduardo Noleto da Nobrega Oliveira, MD, MS, Department of Thoracic Surgery, Barretos Cancer Center, Street Antenor Duarte Viléla, 1331 - Dr. Paulo Prata, Barretos 14784-400, Brazil.
Email: rachidccrr@hotmail.com

compartment and can identify indirect findings such as mediastinal air, fluid collections, and effusions in addition to contrast extravasation.[6-8]

In current clinical practice, imaging evaluation commonly relies on two principal modalities: fluoroscopic esophagography and CT esophagography.[6,9] FE can demonstrate direct contrast extravasation and provides strong confirmatory evidence when a leak is visualized.[4,7] By CTE allows broader assessment of the cervicothoracoabdominal compartment, identifying not only contrast extravasation but also indirect findings such as pneumomediastinum, mediastinal fluid collections, esophageal wall thickening, pleural effusions, and associated complications.[10,11]

Although these modalities are often framed as competing diagnostic tests, their clinical value is not necessarily interchangeable. CTE generally functions as an initial screening and rule-out study, supported by its high sensitivity and ability to define the anatomic extent of disease and related complications.[12,13] FE, in turn, retains an important confirmatory role, particularly when direct visualization of a leak can increase diagnostic certainty and influence subsequent management.[2,14] Therefore, the central clinical question is not simply which modality is more accurate in isolation, but how both tests should be integrated sequentially within a coherent diagnostic pathway.

To address these issues and facilitate decision-making strategies, we conducted a systematic review and meta-analysis of comparative observational studies in adults with suspected esophageal perforation, comparing FE and CTE for diagnostic accuracy. Our aim was to define the pragmatic role of each modality, whether as an initial screening test or a confirmatory test, within a coherent diagnostic algorithm for clinical practice.

Methods

Protocol and Registration

This systematic review and meta-analysis was conducted in accordance with the Enhancing the Quality and Transparency of Health Research (EQUATOR), reporting guidelines, including the Preferred Reporting Items for Systematic Reviews (PRISMA), and the Meta-analysis of Observational Studies in Epidemiology (MOOSE) guidelines. The study protocol was registered on the Open Science Framework (OSF; Registration DOI: 10.17605/OSF.IO/TYU9M). PROSPERO was closed for diagnostic test accuracy registrations at the time of protocol development.

Search Strategy and Study Selection

PubMed, Embase, and Cochrane Library databases were systematically searched from inception to October 26th, 2025. The search strategy was (esophageal perforation* OR oesophageal perforation* OR esophagus perforation* OR Boerhaave syndrome OR esophageal rupture* OR oesophageal rupture* OR esophageal injur* OR oesophageal injur* OR esophageal tear* OR oesophageal tear* OR esophageal leak* OR oesophageal leak* OR esophageal fistula* OR oesophageal fistula* OR pneumomediastinum OR "mediastinal emphysema" OR ((esophageal OR oesophageal) AND (leak OR leakage) AND (surgery OR postoperative OR esophagectomy))) AND ("computed tomography" OR CT OR "CT scan*" OR "CT chest" OR "CT thorax" OR "CT esophagram" OR "CT oesophagram" OR "CT esophagography" OR "CT oesophagography" OR "helical CT" OR "spiral CT" OR "multidetector CT" OR "multi-detector CT" OR MDCT OR "thin-slice CT" OR "thin slice CT" OR "high-resolution CT" OR HRCT OR tomography OR "CT imaging" OR "CT evaluation") AND (fluoroscopy OR fluoroscopic OR esophagram* OR oesophagram* OR esophagography OR oesophagography OR "contrast swallow" OR "barium swallow" OR "gastrografin swallow" OR "water-soluble contrast" OR "upper GI series" OR "upper gastrointestinal series" OR "contrast esophagography" OR "contrast oesophagography" OR (contrast AND radiograph* AND (esophagus OR oesophagus))). References of the included studies and previous systematic reviews and meta-analyses, were also searched aiming for identification and inclusion of additional studies.[15] Two authors (L.M.D. and L.B.G.) independently performed the literature search, imported all retrieved records into Rayyan, a web-based platform for systematic reviews, and conducted the initial screening of studies. After the exclusion of duplicates and titles/abstracts unrelated to the clinical question, the eligibility of each remaining study was assessed based on the full-text review of the articles. Any disagreements were resolved through adjudication by the senior author (R.E.N.N.O.).

Eligibility Criteria

Inclusion in this meta-analysis was restricted to studies that met all of the following criteria: (1) observational studies; (2) enrollment of adult patients (≥18 years), presenting with *suspected esophageal perforation*, including spontaneous, iatrogenic, or traumatic cases; (3) clear reporting of the diagnostic performance of CTE and FE for the diagnosis of esophageal perforation; and (4) availability of sufficient data to construct 2 × 2 contingency tables, including the number of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). Studies were excluded if they (1) were conference abstracts, reviews, animal studies, or case series; (2) were not conducted in the population of interest; (3) did not report diagnostic performance for both

CTE and FE in a way that allowed construction of 2 × 2 contingency tables; or (4) had full text unobtainable.

Data Extraction

Two authors (L.M.D. and M.C.B.B.) independently extracted data from the selected studies using a standardized form. Extracted variables encompassed: general study descriptors (first author, year of publication, study design, and country); patient and clinical characteristics (number of participants, age, sex, and the duration of the follow-up); details of the condition patient condition on presentation, including etiology of perforation (spontaneous, traumatic, or iatrogenic), type of perforation (free vs contained), anatomical location of injury, categorized as cervical, thoracic, or abdominal esophagus and, when available, by esophageal segment (upper, middle, or lower), according to each study's reporting; mortality rate, management strategy and type of the intervention (surgical or endoscopic vs conservative) and diagnostic methodology (reference standard used, and maximum time interval between CTE and FE).

For each diagnostic modality, we also collected detailed test-specific parameters: for CT esophagography, contrast administration protocol, scanner type, and diagnostic accuracy metrics (sensitivity, specificity, positive predictive value [PPV], negative predictive value [NPV], diagnostic odds ratio [DOR], and sample size); and for fluoroscopic esophagography, machine type, contrast administration protocol, and the same diagnostic performance indicators. The numbers of TP, FP, TN, and FN results were obtained or derived from the studies. Any disagreements were resolved by consensus between all authors.

Diagnostic Performance Analysis

Our diagnostic performance analysis of FE and CTE included studies in which both modalities were performed in the same population as an attempt to reduce the clinical (ie, pretest probability of perforation) and methodologic interstudy heterogeneity.

Risk of Bias Assessment

Two independent reviewers (L.B.G. and R.E.N.N.O.) assessed the quality of all eligible studies with the current Quality Assessment of Diagnostic Accuracy Studies (QUADAS)-2, tool (26). This tool is composed of four major domains: patient selection, index test, reference standard, and flow and timing. These domains were then further assessed on the basis of the risk of bias and were rated regarding applicability as "high," "low," "some concerns," or "no information." Disagreements were resolved through consensus between all authors.

Statistical Analysis

The pooled sensitivities and specificities and 95% confidence intervals (95% CIs) were estimated using both univariate and bivariate analyses. First, a univariate random-effects meta-analysis of proportions was performed for sensitivity and specificity using a generalized linear mixed model to generate study-level forest plots. Second, pooled sensitivity, specificity, positive likelihood ratio (PLR), negative likelihood ratio (NLR), and diagnostic odds ratio (DOR) were obtained using a random-effects bivariate model according to the Reitsma method, with PLR, NLR, and DOR derived from the logit-transformed sensitivity and specificity estimates.

Accuracy, positive predictive value (PPV) and negative predictive value (NPV), with 95% CIs, were computed from the modality-specific aggregated 2 × 2 contingency tables (sum of all included studies) using Wilson score intervals. Full and partial areas under the summary receiver operating characteristic (sROC) curves were calculated, and 95% CIs for area under the curve (AUCs) were estimated using the method of Hanley and McNeil.

The summary receiver operating characteristic (sROC) curves and 95% confidence regions were generated directly from the bivariate random-effects Reitsma model. Because the sROC plot is derived from the joint distribution of sensitivity and specificity estimated by this model, all components of the plot—including the summary point, predicted curve, and confidence ellipses—represent exclusively the bivariate pooled estimates.

To directly compare the diagnostic performance of FE and CTE, Z tests were performed on the differences in logit-transformed pooled sensitivities and specificities and on the natural logarithm of the DOR, assuming approximate normality of these transformed measures. Additional Z tests were applied to PPV, NPV, PLR, and NLR derived from the aggregated 2 × 2 data. A two-tailed P value <.05 was considered statistically significant. Between-study heterogeneity was quantified using the I^2 statistic according to the Zhou and Dendukuri method and the Holling unadjusted and sample size–adjusted approaches. All analyses were performed in R (version 4.3.3; R Foundation for Statistical Computing, Vienna, Austria).

Results

Study Selection

As detailed in Figure 1, the initial search identified 2119 records across all databases. After removal of 265 duplicate records and screening of titles and abstracts, 39 studies remained for full-text review inclusion,

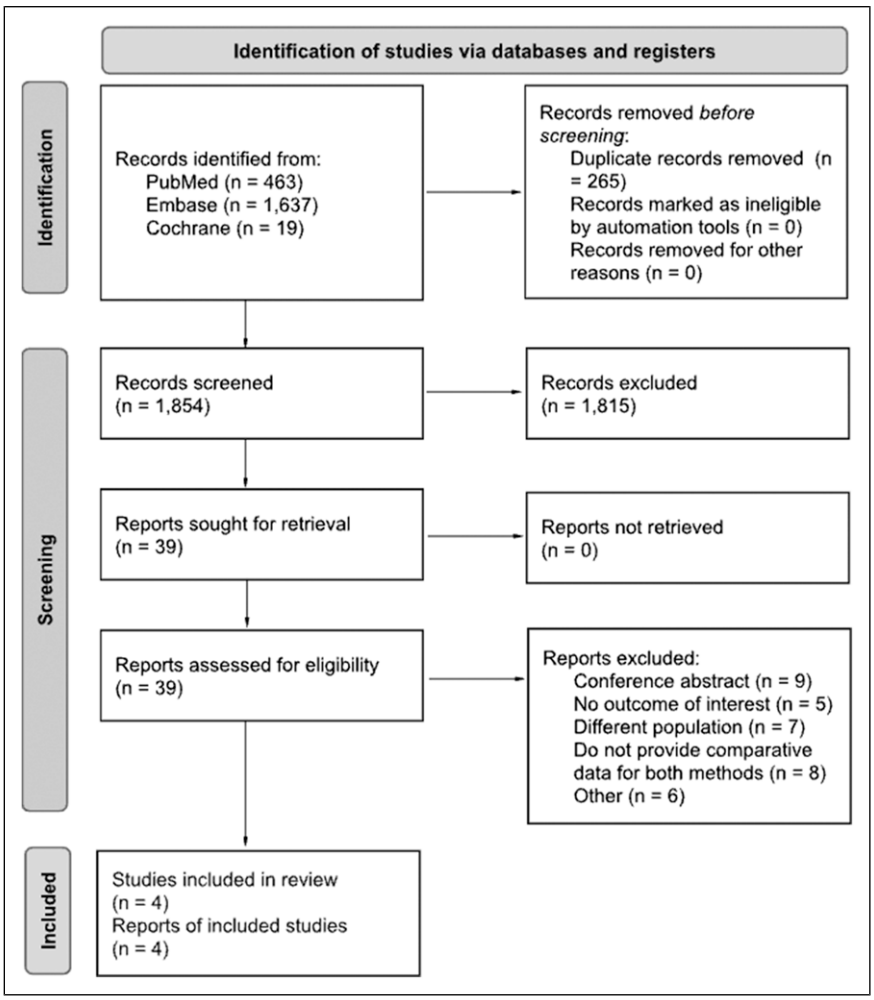


Figure 1. PRISMA flow diagram of study identification, screening, and inclusion

according to prespecified eligibility criteria. Of these, 35 studies were excluded, leaving four retrospective observational studies, published between 2013 and 2020, remaining for inclusion into the analysis. Supplemental Table S2 summarizes the full-text studies excluded at this stage and the reasons for exclusion.

Patient Characteristics

Across the four included studies, a total of 439 patients underwent imaging evaluation for suspected esophageal perforation. Of these, 69 patients (15.6%) were confirmed to have a perforation, while 370 (84.0%) were ultimately classified as non-perforation cases. The screened cohorts were predominantly male, with the proportion of men ranging from 59.2% to 86.7%, and mean or median age across studies ranging from 35.1 to 55 years, reflecting a population of mostly middle-aged adults. When etiologic mechanisms were reported, perforations occurred due to spontaneous, traumatic, or iatrogenic causes, with study-specific proportions varying considerably. The main demographic, etiologic, and imaging protocol characteristics of the included patients are summarized in Table 1. Imaging protocols and injury locations are described in Supplemental Table S3.

Diagnostic Accuracy of FE and CTE

Univariate Analysis. FE demonstrated moderate sensitivity and consistently high specificity across studies. The pooled sensitivity was 78.7% (95% CI: 44.9-94.4; Figure 2A; Table 2F), while specificity was 99.2% (95% CI: 96.7-99.8; Figure 2B; Table 2H). Sensitivity showed moderate variability between studies, whereas specificity remained uniform.

CTE showed an opposite pattern, with very high sensitivity but lower and more variable specificity. The pooled sensitivity was 99.4% (95% CI: 17.9-100.0; Figure 3A; Table 2F), and specificity was 79.8% (95% CI: 64.7-89.5; Figure 3B; Table 2H). Specificity varied substantially across studies, reflecting differences in false-positive rates.

Table 1. Study Characteristics and Across Included Studies

Study	Reference standard	Etiology (%) (S/T/I)	Maximum time between CTE and FE	Male (%)'	Age'	Population screened	Patients with esophageal perforation	Follow-up	
								Patients	Period
Terrazas, 2020	Surgical confirmation or consensus review by a thoracic radiologist, a gastrointestinal radiologist, and a non-cardiac thoracic surgeon	23.4/30.84/ 10.3	24 h, CT first	86 (80.4)	50	110	30	Negative, suspected patients	4 to 6 weeks
Wei, 2020	Surgical confirmation or clinical follow-up	44.4/11.1/ 22.2	27 h, 93,2% performed CT first	61 (59.2)	44.2	103	9	All that didn't undergo procedural intervention	1 week
Awais, 2019	FE	0/53.3/26.7	48 h, CT first	13 (86.7)	55	128	15	NA	NA
Wu, 2013	Surgical confirmation, endoscopic, or FE.	13.3/73.3/6.7	44 h	68 (66.0)	35.1	103	15	Negative diagnosis	4 weeks

Age - mean or median; ' Implies data of the total screened population. CTE = computed tomography; FE = fluoroscopic esophagography; GEJ = gastroesophageal junction; NA = not available; S: Spontaneous; T: Traumatic; I: Iatrogenic.

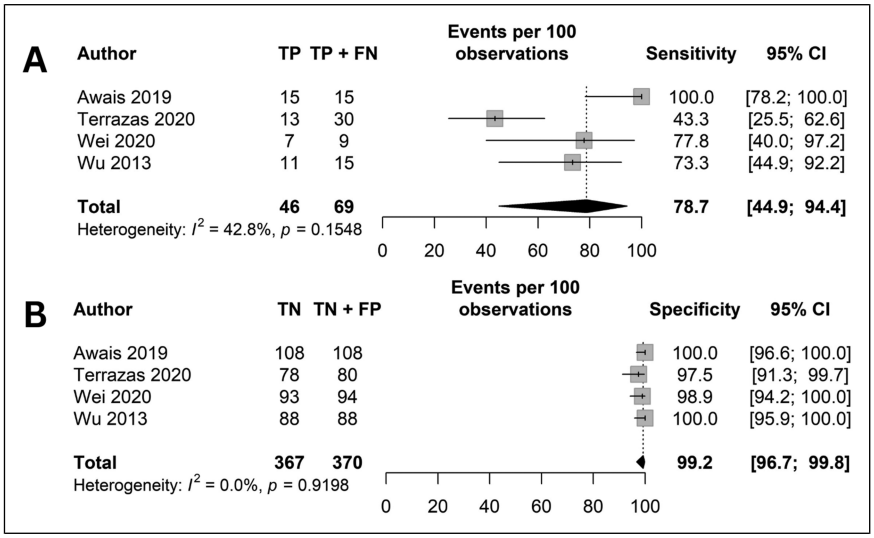


Figure 2. Univariate random-effects meta-analysis of fluoroscopic esophagography (FE). (A) Forest plot of sensitivity; (B) Forest plot of specificity

Bivariate Analysis. FE showed a pooled sensitivity of 69.2% (95% CI: 48.9-84.0; Table 2E) and a markedly high specificity of 98.4% (95% CI: 96.3-99.4; Table 2G). The bivariate random-effects model confirmed FE's strong ability to correctly exclude esophageal perforation, with tight confidence regions on the sROC curve reflecting stable specificity across studies (Figure 4).

CTE demonstrated a higher pooled sensitivity of 81.8% (95% CI: 38.5-97.0; Table 2E) but a lower specificity of 86.7% (95% CI: 67.3-95.4; Table 2G). The wider confidence regions in the sROC plot indicate greater between-study variability, particularly in specificity, consistent with the higher false-positive rates observed in the raw data sets (Figure 4).

Direct Comparison Between FE and CT. Direct statistical comparison using Z tests demonstrated no significant difference between FE and CTE in bivariate pooled sensitivity (P = .469; Table 2E) or specificity (P = .105; Table 2G). FE showed significantly higher PPV than CTE (P < .001; Table 2I) and a substantially greater PLR (P = .007; Table 2K), reflecting its stronger rule-in

Table 2. Diagnostic Performances of Four Studies Using Both FE and CTE as Imaging Modalities for the Evaluation of Esophageal Perforation

Parameter		FE	95% CI	CT	95% CI	*P* value
No. of TP results		46	…	55	…	—
No. of TN results		367	…	282	…	—
No. of FP results		3	…	88	…	—
No. of FN results		27	…	14	…	—
Sensitivity (%)	Bivariate	69.2	48.9-84.0	81.8	38.5-97.0	0.469
	Univariate	78.7	44.9-94.4	99.4	17.9-100.0	—
Specificity (%)	Bivariate	98.4	96.3-99.4	86.7	67.3-95.4	0.105
	Univariate	99.2	96.7-99.8	79.8	64.7-89.5	—
PPV (%)		93.9	83.5-97.9	38.5	30.9-46.6	<0.001
NPV (%)		93.1	90.2-95.2	95.3	92.2-97.2	0.231
PLR		43.3	16.8-111.6	6.2	2.1-17.8	0.007
NLR		0.31	0.17-0.56	0.21	0.04-1.06	0.651
DOR		138.2	39.3-485.8	29.3	3.0-288.1	0.244
AUC		0.978	0.95-1.00	0.911	0.86-0.96	0.013
Accuracy		93.2	90.5-95.2	76.8	72.6-80.5	<0.001

Note. Data are pooled estimates for all four studies (references) of joint FE and CTE in the same population for the evaluation of esophageal perforation. AUC = area under the summary receiver operating characteristic curve, CI = confidence interval, DOR = diagnostic odds ratio, DW = diffusion weighted, FN = false negative, FP = false positive, NLR = negative likelihood ratio, NPV = negative predictive value, PLR = positive likelihood ratio, PPV = positive predictive value, TN = true negative, TP = true positive.

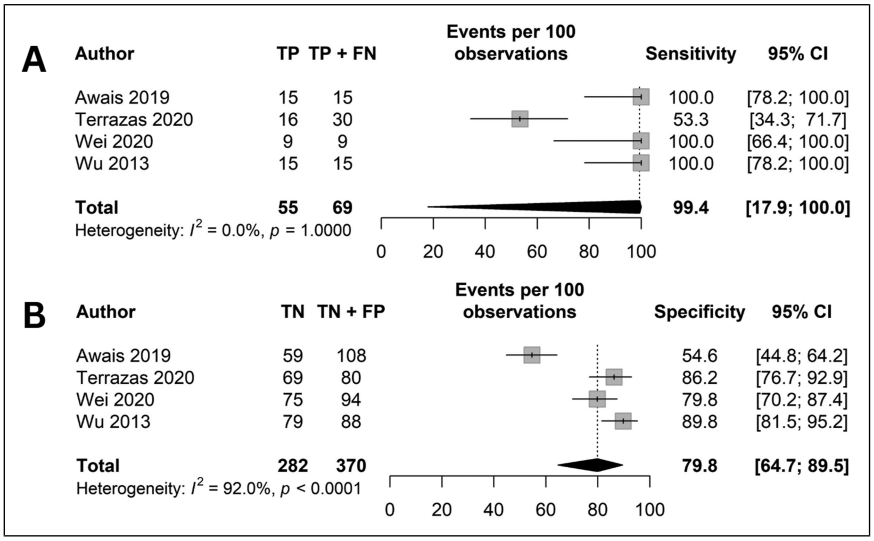


Figure 3. Univariate random-effects meta-analysis of CTE esophagography (CT). (A) FOREST plot of sensitivity; (B) Forest plot of specificity

performance, and accuracy (P < .001; Table 2O). No significant differences were observed for NPV (P = .231; Table 2J), NLR (P = .651; Table 2L), or DOR (P = .244; Table 2M). FE demonstrated a significantly higher AUC than CTE (0.978 vs 0.911; P = .013; Table 2N), indicating superior overall discriminative accuracy.

Risk of Bias Assessment. All QUADAS-2 methodological assessments for the included studies are presented in Figure 5. Overall, patient selection was judged to be at low risk of bias across all four studies, as each used consecutive or clinically representative samples without inappropriate exclusions. In index test assessment, three studies provided adequate information to support a low risk of bias,[7,16,17] whereas one study did not clearly report whether CTE and esophagography interpretations were performed independently of the reference standard.[18] For the reference standard, Awais, Terrazas, and Wu used robust diagnostic confirmation (surgery, endoscopy, or fluoroscopic esophagography), resulting in low risk of

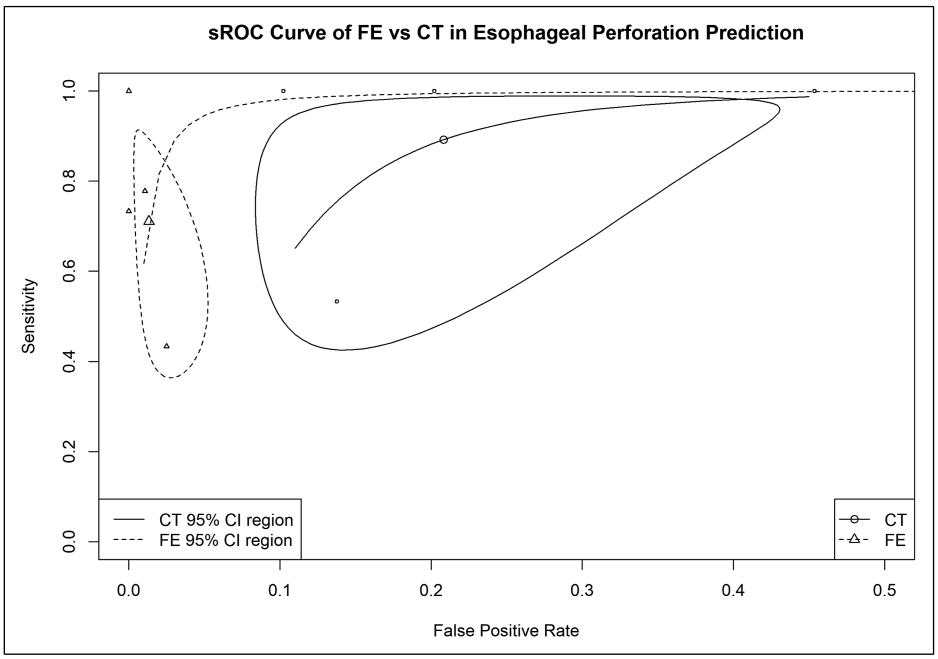


Figure 4. Comparison of bivariate summary receiver operating characteristic (sROC) curves for FE and CT

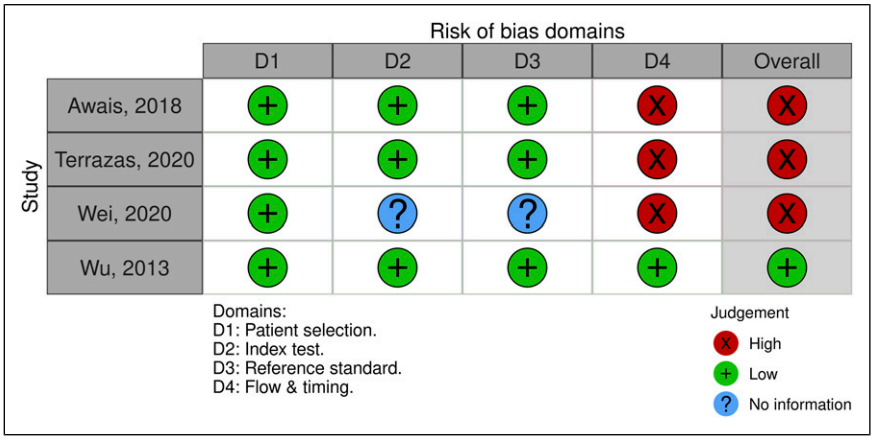


Figure 5. Risk of bias assessment using QUADAS-2

bias[7,16,17]; in contrast, insufficient methodological detail in Wei led to an unclear judgment.

The greatest methodological limitations were observed in flow and timing. Three studies exhibited high risk due to prolonged or inconsistently reported intervals between the index and reference tests,[7,16,18] or because different reference standards were applied across participants. Only one study demonstrated low risk of bias in this domain,[17] with a consistent and clinically appropriate interval (mean 3.1 ± 2.05 hours) and uniform application of the reference standard. These details are fully documented in the Supplemental Table S4.

Discussion

In this systematic review and meta-analysis of four studies including 439 adults with suspected esophageal perforation, we compared the diagnostic performance of FE and CT. Our main findings were as follows: (I) FE demonstrated very high specificity but only moderate sensitivity, confirming its strength as a rule-in test; (II) CTE showed a numerically higher, though not statistically significantly different, sensitivity but lower and more variable specificity, reflecting its greater tendency toward false-positive results; (III) bivariate modeling reinforced these patterns, with FE maintaining superior specificity and CTE offering higher sensitivity but broader confidence regions; and (IV) direct statistical comparisons revealed no significant differences in pooled sensitivity or specificity, although FE exhibited significantly higher PPV, PLR, and AUC, indicating better overall discriminative performance.

This pattern supports a complementary diagnostic strategy: CT, with its numerically higher point-estimate sensitivity, is well suited as an initial test to rule out perforation and to screen for alternative diagnoses in the emergency setting, particularly when rapid triage is essential. FE, with its very high specificity, serves as a confirmatory test when contrast leakage is visualized, increasing post-test probability and strengthening decision-making for invasive interventions.[19-21] The superiority of FE in PPV/PLR and AUC observed in our analysis reinforces its role as the optimal rule-in modality; whereas the wider confidence regions for CTE specificity on the sROC align with higher false-positive variability across studies.

Between-study variability likely reflects differences in imaging protocols, like contrast agents and acquisition techniques for FE; and slice thickness, oral contrast timing, and reconstruction algorithms for CT. Additional variations stem from timing of imaging relative to symptom onset, as early or small, contained leaks may reduce FE sensitivity, and nonspecific indirect CTE signs may inflate false positives. Differences in etiologic mechanisms (spontaneous, iatrogenic, or traumatic), and injury anatomic location (cervical, thoracic, abdominal esophagus) further contribute to heterogeneity in diagnostic performance.[22-24] These factors contextualize the bivariate estimates and explain the wider sROC confidence contours for CT.

The differences observed between the univariate and bivariate pooled estimates were expected and are consistent with the characteristics of our data set. With only four studies and relevant between-study heterogeneity, the bivariate random-effects model, by jointly modeling sensitivity and specificity and accounting for their correlation within each study, tends to reposition the summary point along the trade-off between false positives and false negatives. In contrast, the univariate models treat sensitivity and specificity as independent proportions. In this context, even modest differences in study weights and correlation structures are sufficient to generate slightly different pooled values, without materially changing the overall diagnostic message of our results.[25-28]

This meta-analysis advances the field by providing a standardized, head-to-head comparison of diagnostic

accuracy and clinical utility metrics for FE and CT, using variate diagnostic accuracy metrics, like PPV/NPV, PLR/NLR, diagnostic odds ratio (DOR), and AUC, integrating univariate estimates with a bivariate model and direct statistical comparisons. The finding of a significantly higher AUC for FE, despite no statistically significant differences in sensitivity or specificity in the joint model, clarifies the pragmatic and complementary roles of these modalities in the acute diagnostic pathway.

These findings suggest that the main clinical message of this study should not be interpreted as a direct comparison between two diagnostic modalities, but rather as evidence of complementary functions within a stepwise diagnostic algorithm. CTE, because of its comparatively higher sensitivity and its ability to simultaneously assess the mediastinum, pleural cavity, lung parenchyma, and other cervicothoracoabdominal structures, appears best suited as the initial imaging test in patients with suspected esophageal perforation. In this context, CTE can support early disease exclusion, identify alternative diagnoses, and define the anatomic extent of contamination.[29-31] FE, in contrast, provides strong confirmatory value when direct contrast extravasation is demonstrated, increasing post-test probability and becoming particularly useful when confirmation of an active leak may influence the decision for endoscopic, surgical, or conservative management.[6,32]

Several limitations of this study must be acknowledged. First, the analysis is based on a small number of retrospective studies, introducing potential selection and confounding bias. Second, heterogeneous imaging protocols and reference standards (surgery, endoscopy, and serial CT/clinical follow-up), raise concerns for verification. Additionally, differences in perforation prevalence across studies affect PPV/NPV estimates. Incomplete reporting on symptom-to-test intervals and etiologic subgroups limited the possibility of stratified analysis. In several studies, the time interval between the index tests (CT and FE) and the reference standard was prolonged or inconsistently reported, and different reference standards were applied within the same cohort, which may have differentially influenced the apparent performance of each modality. Finally, the number of studies (<10 studies) was insufficient to assess publication bias. We mitigated these by applying random-effects models, a bivariate framework with sROC, and direct comparisons, while providing qualitative synthesis where appropriate; nonetheless, residual bias cannot be excluded.

Conclusions

In adults with suspected esophageal perforation, CTE and FE should be viewed as complementary rather than interchangeable tests. CTE is best suited as the initial study because of its comparatively higher sensitivity and broad anatomic assessment, whereas FE should be used selectively for confirmation when findings are equivocal or clinical suspicion persists. These findings support a sequential strategy: initial CTE followed by FE when needed.

ORCID iD

Lucas Monteiro Delgado https://orcid.org/0009-0001-7687-3073

Funding

The authors received no financial support for the research, authorship, and/or publication of this article.

Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Supplemental Material

Supplemental material for this article is available online.

References

1. Khaitan PG, Famiglietti A, Watson TJ. The etiology, diagnosis, and management of esophageal perforation. *J Gastrointest Surg*. 2022;26:2606-2615. doi:10.1007/s11605-022-05454-2.
2. Salehpoor A, Thompson W. Imaging of esophageal perforation including in the postoperative patient: a contemporary review. *Abdom Radiol*. 2025;51:650-659. doi:10.1007/s00261-025-05142-0.
3. Chen S, Shapira-Galitz Y, Garber D, Amin MR. Management of iatrogenic cervical esophageal perforations: a narrative review. *JAMA Otolaryngol Head Neck Surg*. 2020;146:488-494. doi:10.1001/jamaoto.2020.0088.
4. Aburumman R, Abdul Razzak F, Kerbage A, et al. Endoscopic management of esophageal perforations: a multicenter study. *Surg Endosc*. 2025;39:6719-6725. doi:10.1007/s00464-025-12090-9.
5. Wang KJ, Alexander EV, Worrell SG. Surgical versus endoscopic management of esophageal perforation. *Curr Opin Gastroenterol*. 2025;41:239-244. doi:10.1097/MOG.0000000000001095.
6. Malach L, Byerly S, Evans CR, et al. Penetrating aerodigestive injuries and the role of computed tomography esophagography. *Am J Surg*. 2025;239:116061. doi:10.1016/j.amjsurg.2024.116061.
7. Terrazas M, Marjon L, Geter M, Schwartz J, Thompson W. Esophagography and chest CT for detection of perforated esophagus: what factors influence accuracy? *Abdom Radiol*. 2020;45:2980-2988. doi:10.1007/s00261-019-02187-w.
8. Petrov RV, Fajardo RA, Bakhos CT, Abbas AE. Peroral endoscopic myotomy: techniques and outcomes. *Shanghai Chest*. 2021;5:14. doi:10.21037/shc.2020.02.02.

9. Chirica M, Kelly MD, Siboni S, et al. Esophageal emergencies: WSES guidelines. *World J Emerg Surg*. 2019;14:26. doi:10.1186/s13017-019-0245-2.
10. Çarkıt S, İpekten F, Karaağaç M, Gök M, Akyuz M. Esophageal perforation management: a single-center experience. *Ulus Travma Acil Cerrahi Derg*. 2024;30:875-882. doi:10.14744/tjtes.2024.26020.
11. Hu Z, Liu D. Mallory-Weiss syndrome. *N Engl J Med*. 2024;390:e49. doi:10.1056/NEJMicm2310882.
12. Jagtiani M, Bhargava R, Mejias C, Watt C, Bhargava P. Can CT esophagogram replace fluoroscopic assessment of suspected esophageal leaks after hours without compromising patient safety? Our 6 year multicenter experience. *Curr Probl Diagn Radiol*. 2026;55:185-187. doi:10.1067/j.cpradiol.2025.10.010.
13. Lee SM, Baek S-E, Lee CW, Kim YC, Kim M-J. Foreign body ingestion: radiologic evaluation, findings, and management. *Korean J Radiol*. 2025;26:638-649. doi:10.3348/kjr.2025.0118.
14. Keogan AG, Rybachok A, Patel S, Latorre-Rodríguez AR, Vittori A, Mittal SK. Diagnostic utility of computed tomography in patients with spontaneous pneumomediastinum evaluated for suspected esophageal perforation. *World J Surg*. 2025;49:3362-3367. doi:10.1002/wjs.70158.
15. Wohlin C. Guidelines for snowballing in systematic literature studies and a replication in software engineering. In Proceedings of the 18th International Conference on Evaluation and Assessment in Software Engineering (EASE '14). Association for Computing Machinery, New York, NY, USA, 2014; Article 38, 1-10. https://doi.org/10.1145/2601248.2601268
16. Awais M, Qamar S, Rehman A, Baloch NU-A, Shafqat G. Accuracy of CT chest without oral contrast for ruling out esophageal perforation using fluoroscopic esophagography as reference standard: a retrospective study. *Eur J Trauma Emerg Surg*. 2019;45:517-525. doi:10.1007/s00068-018-0929-4.
17. Wu C-H, Chen C-M, Chen C-C, et al. Esophagography after pneumomediastinum without CT findings of esophageal perforation: is it necessary? *AJR Am J Roentgenol*. 2013;201:977-984. doi:10.2214/AJR.12.10345.
18. Wei CJ, Levenson RB, Lee KS. Diagnostic utility of CT and fluoroscopic esophagography for suspected esophageal perforation in the emergency department. *Am J Roentgenol*. 2020;215:631-638. doi:10.2214/AJR.19.22166.
19. El-Sourani N. Esophageal perforation: diagnosis, management and decision-making - a retrospective tertiary Centre study. *Turk J Surg*. 2021;37:342-346. doi:10.47717/turkjsurg.2021.5289.
20. Jeong ES, Jung H-K, Byeon JR, et al. Prediction of severity and adverse outcomes associated with perforation in patients presenting to the emergency department with esophageal injury. *Korean J Helicobacter Up Gastrointest Res*. 2023;23:42-51. doi:10.7704/kjhugr.2022.0057.
21. Delgado LM, Meine GC, Santo P, et al. Endoscopic submucosal tunneling techniques versus flexible endoscopic septotomy for Zenker's diverticulum: a systematic review and meta-analysis. *Gastrointest Endosc*. 2025;101:751-761.e30. doi:10.1016/j.gie.2024.11.043.
22. Norton-Gregory AA, Kulkarni NM, O'Connor SD, Budovec JJ, Zorn AP, Desouches SL. CT esophagography for evaluation of esophageal perforation. *Radiographics*. 2021;41:447-461. doi:10.1148/rg.2021200132.
23. Fuhrmann C, Weissenborn M, Salman S. Mediastinal fluid as a predictor for esophageal perforation as the cause of pneumomediastinum. *Emerg Radiol*. 2021;28:233-238. doi:10.1007/s10140-020-01841-x.
24. Tonolini M, Bianco R. Spontaneous esophageal perforation (boerhaave syndrome): diagnosis with CT-esophagography. *J Emergencies, Trauma, Shock*. 2013;6:58-60. doi:10.4103/0974-2700.106329.
25. Chu H, Cole SR. Bivariate meta-analysis of sensitivity and specificity with sparse data: a generalized linear mixed model approach. *J Clin Epidemiol*. 2006;59:1331-1332. doi:10.1016/j.jclinepi.2006.06.011.
26. Reitsma JB, Glas AS, Rutjes AWS, Scholten RJPM, Bossuyt PM, Zwinderman AH. Bivariate analysis of sensitivity and specificity produces informative summary measures in diagnostic reviews. *J Clin Epidemiol*. 2005;58:982-990. doi:10.1016/j.jclinepi.2005.02.022.
27. Leeflang MMG, Deeks JJ, Takwoingi Y, Macaskill P. Cochrane diagnostic test accuracy reviews. *Syst Rev*. 2013;2:82. doi:10.1186/2046-4053-2-82.
28. Miller C, Magarinos J, Akcelik A, Bakhos C, Abbas A, Petrov R. Endoscopic tissue approximation in clinical practice and the OverStitch device: a narrative review. *Ann Esophagus*. 2023;6:21. doi:10.21037/aoe-21-50.
29. Sakurai Y, Takeno A, Tokuyama S, et al. Report of a case of treatment for unresectable esophageal cancer with esophageal perforation and lung abscess. *Gan To Kagaku Ryoho*. 2025;52:1395-1397. PMID: 41546360.
30. Yasrab M, Crawford CK, Chu LC, Kawamoto S, Fishman EK. CT of the esophagus in the ER: what you need to know and what you need to remember. *Emerg Radiol*. 2025;32:447-455. doi:10.1007/s10140-025-02339-0.
31. Nguyen C, Tian C, Samant H. *Esophageal Ulcer. StatPearls, Treasure Island (FL)*. StatPearls Publishing; 2026.
32. Henderson C, Rossanese M, Salmoral A, Murgia D. Surgical management of cricopharyngeal achalasia in spaniels results in favorable outcomes, with persistence or recurrence of mild signs documented in some dogs. *JAVMA (J Am Vet Med Assoc)*. 2026;264:1-6. doi:10.2460/javma.25.06.0415.